

## COM410 Programming in Practice

### B3. Searching

#### CHALLENGE

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##### Timing Search Algorithms

- Generate random arrays of ascending integers of size 1000, 2000, 4000, 8000, 16000, and 32000 elements. For each array the integers should be in the range 1 to the array size \* 10 (e.g. 1K integers in the range up to 10K, 2K integers in the range up to 20K, etc).
  - For each array, generate an array of 1000 search values where the search values are from the same range as the array elements.
  - Measure the time taken by the iterative sequential and recursive binary searches to complete the search for all 1000 search values. Remember to optimize the sequential search for a sorted array.
  - Report the results as shown in the demonstration video.
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Please upload a screenshot of your solution running to the B2 Challenge submission link on Blackboard, by the deadline shown there.